

CIS 613: Software Testing

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Who am I? Dr. Byron DeVries

You can call me:

- Byron
 - Dr. DeVries
 - Professor DeVries
 - Mr. DeVries
 - Mr. Byron
- If you want me to think of you like a middle schooler
- If you want me to think of you like a kindergartener teacher

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Who am I? Dr. Byron DeVries

~10 years teaching Software Engineering at GVSU

~12 Years in industry (*Smiths Aerospace, GE Aviation, Aviage/GE-AVIC*)

I have worked on:

- Requirements through Implementation and Test in:
 - Aircraft Simulation
 - Flight Management Systems (FMS)
- Avionics Cybersecurity
- Process Documentation and Definition for AS9100, CMMI

I have worked as:

- Engineer
- Technical Lead (in Systems, Software, and Verification)
- Project Manager

Most recently: Directed all verification for Boeing 737 and MAX FMS

My bias: high-assurance software development with lots of process.

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Current Research Interests

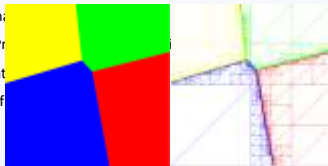
- Software Engineering & Formal Methods
- Evolutionary Computation
- Parallel Processing & Algorithm Design
- Computational Geometry
- Theory of Computation

If this sounds interesting to you, feel free to chat with me or e-mail me during or after the semester.

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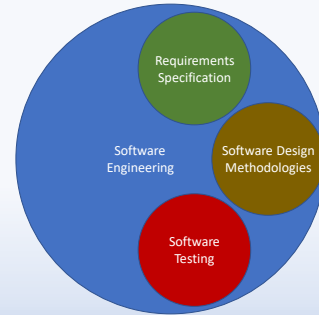
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Course Objectives

- **Describe** the terminology, principles, levels, and types of **software testing**.
- **Apply** both **specification-based** and **code-based techniques** to test a given program.
- **Perform static testing** techniques using reviews and static analysis tools.
- **Understand** the advantages and limitations of **mainline software testing techniques**.
- **Apply software tools** to automate unit, functional, and acceptance testing, and measure code coverage.

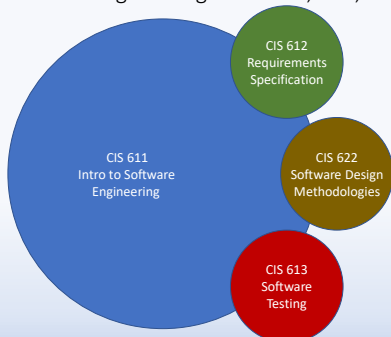
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Software Engineering & Sub-Topics



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CIS 611 Software Engineering & CIS 612, 622, 613



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Syllabus

- **Course Structure**

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Course Structure

- Introduction to Software Testing
- Goals and Objectives
- Specification-Based Testing
- Code-Based Testing
- Exploratory Testing
- Integration Testing
- System Testing
- Software Technical Reviews
- Mutation Testing & Software Complexity
- Test Traceability
- Testing Systems of Systems
- Management & Maintenance of Tests

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Syllabus

- **Course Structure**
- **Assignments**

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Course Assignments

- Class participation
 - Not based *only* on attendance
 - If you are sick, stay home at let me know
 - If the weather is *bad*, stay home and let me know
 - If I think the weather is bad, class will move online. Check your e-mail!
 - *Significant unexcused* absences will impact participation *and* may further impact your grade
- Coursework
 - Homework and written responses to readings
 - Paper / Project of your choice
 - Two assessments (midterm and final)
- Extensions – Not Allowed... Yet.

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Final Exam & Project Schedules

Final Assessment: 04-22
 Project Report: 04-22
 Project Presentation:
 Exam Time: Wednesday, April 29, 6pm-8:50pm

Last Normal Class Day of Semester

Final Exam Day

The Final Assessment is on the Last Day *Normal* Class!

Project Presentations are on the Exam Day!

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Syllabus

- Course Structure
- Assignments
- **Office Hours**

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Office Hours

- Monday, Wednesday: *Allendale* (C-2-209 MAK):
 - 10:50am – Noon
- Wednesday: *Medical Mile*:
 - After Class until 9:30pm or the last person leaves, whichever comes first in DCIH 303.
- Online by appointment!
- E-mail me at: devrieby@gvsu.edu

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Syllabus

- Course Structure
- Assignments
- Office Hours
- **E-mail Policy**
 - No blackboard messages
 - I respond (worst case) by the end of the next Monday, Wednesday, or Friday that is a school day.
 - Check your e-mail frequently. Important information is sent that way regarding class.

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Syllabus

- Course Structure
- Assignments
- Office Hours
- E-mail Policy
- **Early Review**

Note: Grades *tend* to be high or low, but not in-between.

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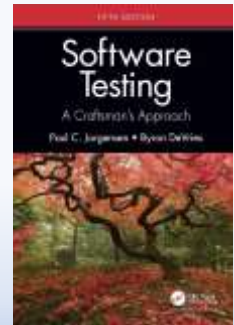
Syllabus

- Course Structure
- Assignments
- Office Hours
- E-mail Policy
- Early Review
- **Pre- and Post-Class Slides**

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Syllabus

- Course Structure
- Assignments
- Office Hours
- E-mail Policy
- Early Review
- Pre- and Post-Class Slides
- **No Required Book!**



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Why no book?



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Fundamental Attribution Error (FAE)

FAE states (roughly) that people:

- Judge others based on their *actions*, but
- Judge themselves based on their *intentions*.

If you promise to assume I am doing my best. Things like:

- Grade fairly,
- Convey information understandably, and
- *Ensure your success in this class.*

Then I promise to assume you are doing your best. Things like:

- Turn in your best effort on time,
- Attend class and participate, and
- *Ensure your success in this class.*

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Project Information

- What *is* it?
 - Project **and** Presentation
- A handful of ideas:
 - A **paper** about software testing.
 - One specific topic,
 - Organize multiple related papers on a single topic
 - A project **about** software testing.
 - Research topics
 - Specific tools
 - Creation, or
 - Exploration / Explanation
 - A project **using** software testing.
 - A project that doesn't have tests, could be
 - Personal
 - Professional
 - Open source

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What is it?

- Project can be *anything* you want it to be that satisfies the following criteria:
 - Must be **related** to software testing,
 - Must **expand** upon and/or **synthesize** the concepts in this class,
 - Must be of sufficient **effort** (*intended to be a ½ semester effort*),
 - Must include a **written and presentation** deliverable,

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Paper

Write a paper on a specific topic. For example:

- Testing a new type of software (e.g., AI)
- Applying a tool to a new domain

~~Summarize or analyze paper(s) from applicable conferences.~~

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Project: **About**

Prove MC/DC coverage is achievable for all non-redundant and non-conflicting sets of conditionals.

Create a tool to perform some kind of testing not easily done.

Explore and explain a tool to perform some kind of testing.

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Project: **Using**

Create a tests for:

- A project from work that doesn't have tests,
- A personal project that could benefit from tests,
- An open source system that could benefit from tests.

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Advice

- If you aren't sure about your idea: Talk to me!
- Don't do *too little*. But don't try to do *too much*!
- Find something **you** are interested in or would benefit **you**.

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10 Questions

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Class Overview

In Class:

- Class Introduction (What we just finished)
- Break (10 minutes)
- Software Testing Introduction
- Break (10 minutes)
- Software Testing Overview
- Background Surveys

After Class:

- Office Hours (until 9:30pm or the last person leaves, whichever comes first)

Homework:

- 01-21: Syllabus Quiz
- 01-21: Unit Testing Frameworks
- **Optional:** Read Software Testing Ch 1, 5, and 6

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